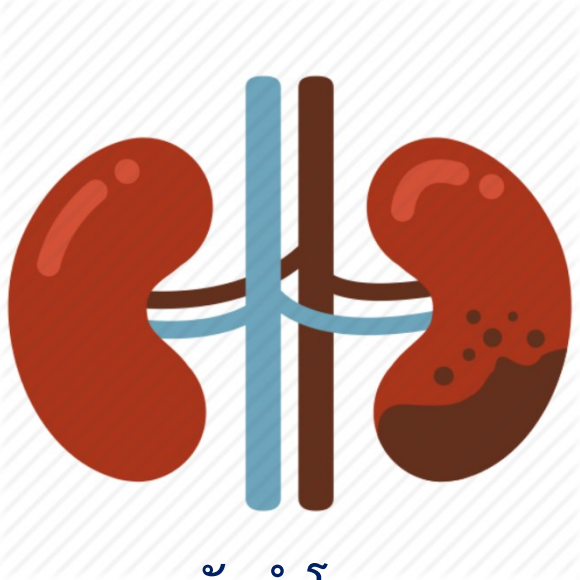
ขนาดยาในผู้ใหญ่ที่มีการทำงานของไตบกพร่อง



จัดทำโดย

งานวิชาการเภสัชสนเทศ กลุ่มงานเภสัชกรรมโรงพยาบาลมหาสารคาม

พ.ศ. 2568

ยาฆ่าเชื้อแบคทีเรีย

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Penicillins									
Amoxicillin		500-1,000 mg po q8h (25-50 MKD divided q8h or 80-100 MKD divide q8-12h for AOM)	No dosage adjustment		250-500 mg po q12h	250-500 mg po q24h	250-500 mg po q24h (+ extra dose AD)	250-500 mg po q12h	250-500 mg po q8-12h
Amoxicillin/ Clavulanate (Augmentin®)	**Oral**	500/125 mg po q8h or 875/125 mg po q12h	No dosage adjustment		250-500 mg (amox) poq12h	250-500 mg (amox) q24h	250-500 mg (amox) po q24h (+ extra dose AD)	No data	No data
	IV	(1,000 mg/200 mg) 1.2 g IV q8h	No dosage adjustment		1.2 g load then 600 mg IV q12h	1.2 g load then 600 mg IV q24h	1.2 mg load then 600 mg IV q24h (+ extra dose AD)	No data	No data
Ampicillin		1-2 g IV q4-6h (max 12 g/day)	No dosage adjustment	1-2 g IV q6-8h	1-2 g IV q8-12h	1-2 g IV q12h	1-2 g IV q12h (give one of the dialysis day doses AD)	500-1,000 mg IV q12h	1-2 g IV q8-12h
Ampicillin/Sulbactam (Unasyn®: 2 g/1 g)		1.5-3 g IV q6h ***not for A.baum***	No dosage adjustment		CrCl 29-15 1.5-3g IV q12h	CrCl <15 1.5-3 g IVq24h	1.5-3 g IV q24h (give AD on dialysis day)	3 g IV q24h	1.5-3 g IV q8-12h
		9-12 g Sulbactam/day ***A.baum MDR***	CrCl ≥30 **(Sulb. 10 g/day)** Unasyn 3 g IV q6h + Sulperazone 4g IV q8h		CrCl 29-15 **(Sulb. 4.5-6 g/day)** Unasyn 3 g IV q6h	CrCl <15 **(Sulb. 3 g/day)** Unasyn 3 g IV q8h	No data	No data	No data
Penicillin G		0.5-4 mU IV q4h	No dosage adjustment	0.5-4 mU IV q8h		0.5-4 mU IV q12h	0.5-4 mU IV q12h (give one of the dialysis day doses AD)	0.5-4 mU IV q12h	1-4 mU IV q6-8h
Cloxacillin*		2 g q4-6h (max: 12 g/day)	No dosage adjustment						No data

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Dicloxacillin		250-500 mg po q6h	No dosage adjustment						No data
Penicillin V		250-500 mg po q6-8h	No dosage adjustment				250-500 mg po q6-8h (give at least one dose AD)	No dosage adjustment	No data
Piperacillin/ Tazobactam (4g/0.5g)	**Pseudomonas dose**	4.5 g IV q6h (over 30 min)	CrCl>40 No dosage adjustment	CrCl 40-20 3.375 g IV q6h	CrCl<20 2.25 g IV q6h		2.25 g IV q8h (+ extra 0.75 g AD)	2.25 g IV q8h	MIC≤16: 3.375 g IV q6h MIC 16-64: 4.5 g (over 4 h) IV q8h
	Non-Pseudomonas dose	3.375 g IV q6h (over 30 min)	CrCl>40 No dosage adjustment	CrCl 40-20 2.25 g IV q6h	CrCl<20 2.25 g IV q8h		2.25 g IV q12h (+ extra 0.75 g AD)	2.25 g IV q12h	2.25 g IV q6h
Cephalosporins									
Cefazolin		1-2 g IV q8h	No dosage adjustment	0.5-2 g IV q8-12h		0.5-1 g IV q24h	0.5-1 gIV q24h (dose AD on dialysis days)	0.5 g IV q12h	1-2 g IV q12h
Cefdinir		300 mg po q12h	No dosage adjustment		300 mg po q24h		300 mg po q24h (give AD on dialysis days)	300 mg po q24h	No data
Cefixime		400 mg po q24h	No dosage adjustment	300 mg po q24h		200 mg po q24h	200 mg po q24h (give AD on dialysis days)	200 mg po q24h	No data
Cefoperazone/ Sulbactam (0.5 g/0.5 g)		1-2 g (Cefo) IV q12h ***not for A.baum***	No dosage adjustment		CrCl 30-15 1 g IV q12h	CrCl <15 500 mg IV q12h	500 mg IV q12h+ (give one of the dialysis day doses AD)	No data	No data
		9-12 g (Sulb.)/day ***A.baum MDR***	CrCl ≥30 **(Sulb. 10 g/day)** Unasyn 3 g IV q6h + Sulperazone 4g IV q8h		CrCl 29-15 **(Sulb. 4.5-6 g/day)** Unasyn 3 g q6h	CrCl <15 **(Sulb. 3 g/day)** Unasyn 3 g q8h	No data	No data	No data
Cefotaxime		2 g IV q8h	2 g IV q8-12h	2 g IV q12-24h		2 g IV q24h	2 g IV q24h (+ extra 1 g AD)	0.5-1 g IV q24h	2 g q12-24h

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
		90-50	49-30	29-10	<10			
Ceftazidime	2 g IV q8h	2 g IV q8-12h	2 g IV q12-24h		2 g IV q24h	0.5-1 g q24h (AD on dialysis days)	**Non-PD infect:** 1.5-2 g IP q24-48h **PD infect:** 3 g IP then 1-2 g IP q24h or 2 g IP q48h	1-2 g q8-12h
Ceftazidime/Avibactam (2 g/0.5 g)	**Adult:** 2.5 g IV q8h	No dosage adjustment	1.25 g IV q8h	CrCl 30-16 938 mg IV q12h	CrCl 15-6 938 mg q24h	938 mg IV q48h (dose AD on dialysis days)	No data	No data
	Newborn-Child<2yr: 62.5 MK IV q8h	No dosage adjustment	31.25 MK q8h	CrCl 30-16 23.75 MK q12h	CrCl 15-6 23.75 MK q24h	23.75 MK q48h	No data	No data
Ceftriaxone	1-2 g IV q12-24h	No dosage adjustment						
Cephalexin	500-1,000 mg po q6h	No dosage adjustment	250-1,000 mg po q8-12h		250-1,000 mg po q24-48h	250-500 mg po q12-24h (give AD on dialysis days)	250-500 mg po q12-24h	No data
Carbapenems								
Ertapenem	1 g IV q24h	No dosage adjustment		0.5 g IV q24h		0.5 g IV q24h (+ extra 150 mg AD, if dose ≤6 h before HD) or 0.5 g 3 time/wk AD	0.5 g IV q24h	0.5-1 g IV q24h
Imipenem/Cilastin	**Usual dose:** 500 mg IV q6h or 1 g IV q8h	CrCl 90-60 500 mg IV q6h	CrCl 60-30 500 mg IV q8h	CrCl 30-15 500 mg IV q12h	CrCl <15 No data	500 mg IV q12h (Administration AD and at intervals timed from the end of that HD session)	125-250 mg IV q12h	0.5-1 g IV q12h
	Intermediate susceptibility: 1 g IV q6h	CrCl 90-60 750 mg q8h	CrCl 60-30 500 mg q6h	CrCl 30-15 500 mg IV q12h	CrCl <15 No data			

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Meropenem		Mild-moderate: 1 g IV q8h	No dosage adjustment	CrCl 50-25 1 g IV q12h	CrCl 25-10 0.5 g IV q12h	CrCl <10 0.5 g IV q24h	0.5 g IV q24h (give AD on dialysis days)	0.5 g IV q24h	750 mg IV q8h
		Severe: 2 g IV q8h	No dosage adjustment	CrCl 50-25 2 g IV q12h	CrCl 25-10 1 g IV q12h	CrCl <10 1 g IV q24h	1 g IV q24h (give AD on dialysis days)	1 g IV q24h	1.5 g IV q8h
Macrolides									
Azithromycin	Oral	500 mg day 1 then 250 mg OD day 2-5	No dosage adjustment						
	IV	500 mg IV OD							
Clarithromycin		500 mg po q12h	No dosage adjustment	500 mg po q12-24h		500 mg po q24h	500 mg po q24h (give AD on dialysis days)	500 mg po q24h	500 mg po q12-24h
Erythromycin		250-500 mg po q6h	No dosage adjustment						
Roxithromycin*		150 mg po q12h	No dosage adjustment [limited data]						
Aminoglycosides									
Amikacin		7.5 mg/kg IV q12h	No dosage adjustment	7.5 mg/kg IV q24h		7.5 mg/kg IV q48h	7.5 mg/kg IV q48h (+ extra 3.75 mg/kg AD)	Peritonitis: 2 mg/kg IP q24h	7.5 mg/kg IV q24h
		15 mg/kg IV q24hr	CrCl >80 15 MK q24h	CrCl 80-60 12 MK q24h	CrCl 60-40 7.5 MK q24h	CrCl 40-30 4 MK q24h	CrCl 30-20 7.5 MK q48h	CrCl 20-10 4 MK q48h	CrCl <10 3 MK q72h	3 mg/kg IV q72h and AD	No data	No data
Gentamicin		1.7-2.0 mg/kg IV q8h	No dosage adjustment	1.7-2.0 mg/kg IV q12-24h		1.7-2.0 mg/kg IV q48h	1.7-2.0 mg/kg IV q48h (+extra 0.85-1.0 MK AD)	Peritonitis:0.6 mg/kg IP once daily	1.7-2.0 mg/kg IV q24h
		5.1 mg/kg IV q24hr	CrCl >80 5.1 MK q24h	CrCl 80-60 4 MK q24h	CrCl 60-40 3.5 MK q24h	CrCl 40-30 2.5 MK q24h	CrCl 30-20 4 MK q48h	CrCl 20-10 3 MK q48h	CrCl <10 2 MK q72h	2 mg/kg IV q72h and AD	No data	No data

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Quinolones									
Ciprofloxacin	**Oral**	250-750 mg po q12h	No dosage adjustment	250-500 mg po q12h	CrCl 29-5 250-500 po mg q24h		250-500 mg po q24h (give AD on dialysis days)	250-500 mg po q24h	250-500 mg po q12h
	IV	200-400 mg IV q8-12h	No dosage adjustment		CrCl 29-5 200-400 mg IV q18-24h		200-400 mg IV q24h (give AD on dialysis days)	200-400 mg IV q24h	200-400 mg IV q12h
Levofloxacin		750 mg po/IV q24h	No dosage adjustment	CrCl 49-20 750 mg q48h	CrCl<20 750 mg once, then 500 mg q48h		750 mg once, then 500 mg q48h		
Norfloxacin		400 mg po q12h	No dosage adjustment		400 mg po q24h		400 mg po q24h		Not applicable
Ofloxacin		200-400 mg po q12h	No dosage adjustment	200-400 mg po q24h		200 mg po q24h	200 mg po q24h (give AD on dialysis days)	200 mg po q24h	200-400 mg po q24h
Sitafloxacin		100 mg po BID	No dosage adjustment	100 mg q24h	100 mg q48h	No data			
Some other antibacterials									
Dapsone		50-300 mg po OD	No dosage adjustment						
Doxycycline		100 mg po q12h	No dosage adjustment						
Fusidic acid		250-750 mg po q8-12h	No dosage adjustment				No data		
Clindamycin*	**Oral**	150-450 mg q6-8h	No dosage adjustment						
	IV	600-900 mg IV q8h (max 4.8 g/day)							
Metronidazole		7.5 mg/kg IV/po q6-8h (max 4 g/day)	No dosage adjustment			7.5 mg/kg q12h	7.5 mg/kg q12h (give one of the dialysis day dose AD)	7.5 mg/kg q6h	

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Fosfomycin	Meningitis	16 - 24 g/day IV แบ่งให้ 3-4 ครั้ง (8 g IV q8h)	CrCl 31-40 12 - 18 g/day (4 g IV q8h)	CrCl 21-30 10 - 14 g/day (4 g IV q8h)	CrCl 11-20 6 - 10 g/day (2 g IV q8h)	CrCl <10 3 - 5 g/day (1 g IV q8h)	2 g IV q48h (at end of each dialysis session)	No data	No data
	Osteomyelitis	12 -24 g/day IV แบ่งให้ 2-3 ครั้ง (4 g IV q8h)	8 - 17 g/day (4 g IV q8h)	7 - 14 g/day (4 g IV q8h)	5 - 10 g/day (2 g IV q8h)	2 - 5 g/day (2 g IV q12h)	2 g IV q48h (at end of each dialysis session)	No data	No data
	HAP	12 -24 g/day IV แบ่งให้ 2-3 ครั้ง (4 g IV q8h)	8 - 17 g/day (4 g IV q8h)	7 - 14 g/day (4 g IV q8h)	5 - 10 g/day (2 g IV q8h)	2 - 5 g/day (2 g IV q12h)	2 g IV q48h (at end of each dialysis session)	No data	No data
	Complicated UTI	12 - 18 g/day IV แบ่งให้ 2-3ครั้ง (4 g IV q8h)	8 - 12 g/day (4 g IV q12h)	7 - 10 g/day (4 g IV q12h)	5 - 6 g/day (2 g IV q8h)	2 - 3 g/day (1 g IV q12h)	2 g IV q48h (at end of each dialysis session)	No data	No data
Colistin* (Polymyxin E)		LD 300 mg IV then 2.5-5 mg/kg/day (Max :5 mg/kg/day)	CrCl >80 LD 300 mg then 150 mg q8-12h	CrCl 80-40 LD 300 mg then 150 mg q12h	CrCl 40-20 LD 300 mg then 100 mg q12h	CrC<20 LD 300 mg then 150 mg q24h	**LD 300 mg then; Non-dialysis day**: 150 mg q24h, **Dialysis day:** 200 mg after dialysis	LD 300 mg then 100 mg q24h	Add 13 mg/h of CRRT to the baseline dose of 150 mg q24h
Sulbactam		Adult: 9-12 g /day divided q6-12h	CrCl ≥30 9-12 g/day		CrCl 29-15 4.5-6 g/day	CrCl <15 2.5-3 g/day	No data	No data	No data
		Children: 200 MKD Neonate: 100-200 MKD	CrCl ≥30 200 MKD divided q6-8h		CrCl 29-15 150 MKD divided q8-12h	CrCl <15 75-100 MKD divided q24h	75-100 MKD	75-100 MKD	No data
Sulfamethoxazole/ Trimethoprim (400 mg/80 mg)		TMP 5-20 mg/kg/day IV, po (divided q6-12h)	No dosage adjustment		TMP 5-10 mg/kg/day (divided q12h)	Not recommended; but if use: TMP 5-10 mg/kg q24h (give dialysis day dose AD)			5 mg/kg po/IV q8h

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
		90-50	49-30	29-10	<10			
Tigecycline	LD 100-200 mg IV then 50-100 mg q12h	No dosage adjustment						
Vancomycin	LD 20-30 mg/kg, use ABW, consider for serious infection **(Max loading dose = 3 g/dose)**							
	15-20 MK q8-12hr (Use actual BW; No single dose over 2 g.)	CrCl 100-51 15-20 MK q12hr	CrCl 50-20 15-20 MK q24hr		CrC<20 15-20 MK q48hr or longer	**Low perm membrane:** MD 7.5 MK AD **High perm membrane:** MD 10 MK AD **>>ให้ยาเฉพาะวันที่ฟอกไต<<**	q48-96h or 1 g q7days 7.5-10 mg/kg q12h	(effluent 20-25 mL/kg/h)
อ้างอิง: The Sanford Guide to Antimicrobial Therapy, 2023.* Drug information handbook 25th edition, ** Micromedex, + Leaflet ceftomax-s, ++ Leaflet FDA AD = after hemodialysis, LD = Loading dose, MD = Maintenance dose, MK = mg/kg								

ยาฆ่าเชื้อไวรัส

ชื่อยา		Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
			90-50	49-30	29-10	<10			
Acyclovir	Oral	200 mg poq4h	No dosage adjustment			200 mg po q12h	200 mg po q12h (give extra dose AD)	200 mg po q12h	No data
		400 mg po q12h	No dosage adjustment						
		800 mg po q4h	No dosage adjustment		CrCl 25-10 800 mg q8h	800 mg po q12h	800 mg po q12h (give extra dose AD)	800 mg po q12h	
	IV	5-12.5 mg/kg q8h (Obesity; Use AdjBW)	No dosage adjustment	5-12.5 mg/kg IV q12-24h		2.5-6.25 mg/kg IV q24h	2.5-6.25 mg/kg IV q24h (give AD on dialysis days)	2.5-6.25 mg/kg IV q24h	
Oseltamivir		**Prophylaxis:** 75 mg OD x 10 day	CrCl 90-60 No dosage adjustment	CrCl 59-30 30 mg OD	30 mg EOD	No data	No data	30 mg weekly (after a dialysate exchange)	No data
		Treatment: 75 mg BID x 5day	CrCl 90-60 No dosage adjustment	CrCl 59-30 30 mg BID	30 mg OD	Not recommended	30 mg after each dialysis; no drug on non-dialysis days	30 mg x1 dose (after a dialysate exchange)	No data
Favipiravir		35 mg/kgq12h day 1, then 15 mg/kg q12h x5-10 day	No dosage adjustment						No data
Molnupiravir		4 tab BID x5 day	No dosage adjustment						No data
Remdesivir		LD 200 mg IV day 1, then 100 mg q24h	No dosage adjustment		No data (Not recommended)		No data	No data	No data
Ganciclovir	induction	5 MK IV q12h	CrCl 69-50 2.5 MK q12h	CrCl 49-25 2.5 MK 242h	CrCl 24-10 1.25 MK 242h	1.25 MK 3x/week	1.25 MK 3x/week (give AD on dialysis days)	1.25 MK 3x/week	2.5 MK q24h
	maintenance	5 MK IV q24h	2.5-5 MK IV q24h	0.625-1.25 MK IV q24h		0.625 MK 3x/week	0.625 MK 3x/week (give AD on dialysis days)	0.625 MK 3x/week	No data
อ้างอิง: The Sanford Guide to Antimicrobial Therapy, 2023									

ยาฆ่าเชื้อรา

ชื่อยา	Normal dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
		≥50	49-10	<10			
Amphotericin B	0.4-1 mg/kg/day (Obesity; Use LBW)	No dosage adjustment					
Fluconazole	100-400 mg IV/po q24h **Crypto meningitis:** 800-1,200 mg/day	No dosage adjustment	50% reduction		50% reduction on non-dialysis days, full dose AD on dialysis day	50% reduction	Full dose
Flucytosine	25 mg/kg po q6h	CrCl 40-21 25 mg/kg q12h	CrCl 20-10 25 mg/kg q24h	25 mg/kg q48h	25-50 mg/kg q8-72h (dose AD)	No data	No data
Griseofulvin	500-1,000 mg OD หรือ divide bid	No dosage adjustment					
Itraconazole	100-200 mg po q12h	No dosage adjustment		50-100 mg q12h	100 mg q12-24h		100-200 mg q12h
Nystatin	500,000 U qid	No dosage adjustment			No data		
Voriconazole	**BW >40kg:** 400 mg po q12h x 2 dose, then 200 mg po q12h **BW <40 kg**: 200 mg po q12h x 2 doses, then 100 mg po q 12 h	No dosage adjustment					
อ้างอิง: The Sanford Guide to Antimicrobial Therapy, 2020							

ยาต้านไวรัสHIV

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD*	CRRT
		≥50	49-30	29-10	<10			
NRTIs								
3TC (Lamivudine)	(HIV) 300 mg q24h	No dosage adjustment	50-150 mg q24h		25-50 mg q24h	25-50 mg q24h วันฟอกไตให้ยาหลังฟอกไต	25-50 mg q24h	100 mg 1st day, then 50 mg q24h
	(HBV) 150 mg q24h	No dosage adjustment	50 mg q24h or 100 mg EOD	CrCl 30-15 25 mg q24h or 100 mg q72h	CrCl <15 100 mg q7day	100 mg q7day วันฟอกไตให้ยาหลังฟอกไต	No data	No data
FTC (Emtricitabine)	200 mg q24h	No dosage adjustment	200 mg q48h	200 mg q72h	200 mg q96h	200 mg q96h วันฟอกไตให้ยาหลังฟอกไต	No data	No data
TAF (Tenofovir)	25 mg q24h	CrCl ≥15 No dosage adjustment			CrCl <15 Not recommended	25 mg q24h วันฟอกไตให้ยาหลังฟอกไต	No data	No data
TDF (Tenofovir)	300 mg q24h	No dosage adjustment	300 mg q48h	300 mg q72-96h	No data	300 mg q7day วันฟอกไตให้ยาหลังฟอกไต	No data	No data
TDF/FTC	300/200 mg q24h	No dosage adjustment	1 tab q48h	ไม่แนะนำ				
TAF/FTC	25/200 mg q24h	No dosage adjustment		ไม่แนะนำ		Normal dose (close monitoring)	Normal dose (close monitoring)	Normal dose (close monitoring)
AZT (Zidovudine)	300 mg q12h	No dosage adjustment			100 mg q8h	100 mg q8h (dose AD on dialysis days)	100 mg q8h	300 mg q12h
AZT/3TC (Zilavir®)	300/150 mg q12h	No dosage adjustment	No recommended (ควรใช้ยาแยกเม็ดและปรับขนาดยาแต่ละชนิดตามการทำงานของไต)					
ABC (Abacavir)	300 mg q12h or 600 mg q24h	No dosage adjustment						
ABC/3TC	600/300 mg q24h	No dosage adjustment	ให้ใช้ยาเม็ดแยกแทนโดยปรับขนาด 3TC ตาม CrCl					

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD*	CRRT
		≥50	49-30	29-10	<10			
NNRTIs								
EFV (Efavirenz)	600 mg q24h hs	No dosage adjustment						
NVP (Nevirapine)	200 mg q12h	No dosage adjustment						
RVP (Rilpivirine)	25 mg q24h	No dosage adjustment		No dosage adjustment; use with caution				
EFV/TDF/FTC	600/300/200 mg q24h	No dosage adjustment	ไม่แนะนำให้ใช้ ควรใช้ยาแยกเม็ดและปรับขนาดยาแต่ละชนิดตามการทำงานของไต					
Protease Inhibitor (PIs)								
ATV (Atazanavir)	400 mg q24h หรือATV/RTV 300/100 mg q24h	No dosage adjustment					No data	No data
DRV/r (Darunavir/ritonavir)	800/100 mg q24h (naïve) หรือ 600/100 mg q12h	No dosage adjustment					No data	No data
DRV/cobi (Darunavir/cobicistat)	800/150 mg q24h	CrCl < 70 ไม่ควรใช้ยานี้ร่วมกับ TDF				No data	No data	No data
LPV/r (Lopinavir/ritonavir)	400/100 mg q12h	No dosage adjustment					No data	No data
Integrase inhibitors								
DTG (Dolutegravir)	50 mg q24h	No dosage adjustment					No data	No data
RAL (Raltegravir)	400 mg q12hหรือ 1,200 mg q24h	No dosage adjustment					No data	No data
DTG/3TC/TDF (Acriptega®)	50/300/300 mg 1tabq24h	No dosage adjustment	Not recommended			Not recommended		

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD*	CRRT
		≥50	49-30	29-10	<10			
DTG/FTC/TAF (Kocitaf®)	50/200/25 mg 1tab q24h	No dosage adjustment		Not recommended				
DTG/ABC/3TC (Triumeq®)	50/600/300 mg 1tab q24h	No dosage adjustment		Do not use				

อ้างอิง: แนวทางการตรวจวินิจฉัย รักษาและป้องกันการติดเชื้อเอชไอวี ประเทศไทย ปี 2564/2565, * The Sanford Guide to Antimicrobial Therapy, 2020.

ยารักษาวัณโรค

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-50	49-10	<10			
First line drugs								
Isoniazid	4-6 MKD	300mg/day	No dosage adjustment					
Rifampicin	8-12 MKD	600 mg/day	No dosage adjustment					
Pyrazinamide	20-30 MK po q24h	2 g/day	No dosage adjustment	CrCl<30, hemodialysis: 20-30 mg/kg/day - 3 times/wk วันฟอกไตให้ยาหลังฟอกไต			25 mg/kg q24h	
Ethambutol	15-20 MK po q24h	1,200 mg/day	No dosage adjustment	CrCl<30, hemodialysis: 15-20 mg/kg/day - 3 times/wkวันฟอกไตให้ยาหลังฟอกไต			15 MK q48h	15-25 MK q24h
Drugs for MDR TB								
Clofazimine	100 mg po q24h	100 mg/day	No dosage adjustment					No data
Cycloserine	10-15 MKD or 250-500 mg q12h	1,000 mg/day	No dosage adjustment	250-500 mg q12-24h	500 mg q48h or 3 times/wk	250 mg OD or 500 mg 3 times/wk วันฟอกไตให้ยาหลังฟอกไต	No data	No data
Ethionamide	15-20 MKD or 500 mg q12h	1 g/day	No dosage adjustment		50% reduction or 250 mg q12h	50% reduction or 250 mg q12h	50% reduction or 250 mg q12h	500 mg q12h
Prothionamide	15-20 MKD	1 g/day	No data					
Moxifloxacin	400 mg po q24h	400 mg/day	No dosage adjustment					
Levofloxacin	15 MKD or 500-1,000 mg po q24h	1.5 g/day	No dosage adjustment	CrCl 49-20 750 mg q48h	CrCl <20 750 mg once, then 500 mg q48h	750 mg once, then 500 mg q48h	750 mg once, then 500 mg q48h	750 mg once, then 500 mg q48h

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-50	49-10	<10			
Linezolid	**TB:** 600 mg q24h **MRSA, VRE:** 600 mg q12h	1.2 g/day	No dosage adjustment					
Bedaquiline	400 mg/day first 2 weeks, then 200 mg/day 3 times/week 22 weeks (total 24 weeks)	400 mg/day	No dosageadjustment					
อ้างอิง: The Sanford Guide To Antimicrobial Therapy, 2020; แนวทางการควบคุมวัณโรค ประเทศไทยพ.ศ. 2564; Drug Information Handbook , 25th edition								

ยาลดความดันโลหิต

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-51	50-10	<10			
Amiloride	5-10 mg OD	40 mg/day	No dosage adjustment	Administer 50% of normal dose	ไม่แนะนำ	No data		Administer 50% of normal dose
Furosemide	200-300 mg/dose q12-24h	Oral: 600 mg/day	No dosage adjustment					
Hydrochlorothiazide	12.5-200 mg OD	200 mg	No dosage adjustment		CrCl<30 ไม่แนะนำ	No data		
Mannitol	50-100 g IV for 15%-25% sol	150 g	No dosage adjustment		Contraindication	No data		
Spironolactone	25-400 mg daily (varyตาม indication)	400 mg/day	50% of normal dose		Use with caution	Use with caution		50% of normal dose
Hydralazine	25-50 mg q8h	300 mg/day	No dosage adjustment					
Minoxidil	5-30 mg bid	60 mg/day	No dosage adjustment					
Methyldopa	250-800 mg q8h	2,400 mg/day	No dosage adjustment	250-800 mg q8-12h	250-800 mg q12-24h	250-800 mg q12-24h วันฟอกไตให้ยาหลังฟอก	250-800 mg q12-24h	250-800 mg q8-12h
Doxazosin	1-16 mg OD	16 mg/day	No dosage adjustment					
Enalapril	5-20 mg q12-24h	40 mg/day	No dosage adjustment	50-100% จากขนาดยาปกติ	25% จากขนาดยาปกติ	No data		
Ramipril	2.5-10 mg OD	10 mg/day	CrCl >20 No dosage adjustment		CrCl <20 Initial dose 1.25 mg daily	No data		
Azilsartan	40-80 mg OD	80 mg/day	CrCl >20 No dosage adjustment		CrCl <20 Initial dose 20 mg daily	Initial dose 20 mg daily		
Candesartan	2-32 mg OD	32 mg/day	CrCl >20 No dosage adjustment		CrCl <20 Initial dose 2 mg daily	Initial dose 2 mg		

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-51	50-10	<10			
Losartan	**HT:** 25-100 mg OD **HF:** 12.5-150 mg OD	150 mg/day	CrCl >20 No dosage adjustment		CrCl <20 Initial 25 mg daily	Initial 25 mg daily		
Sacubitril/Valsartan	100 mg (49/51) BID	200 mg BID	eGFR >30 No dosage adjustment		eGFR <30 initial 50 mg (1/2 tab) BID	No data		
Captopril	6.25-50 mg q8-12h	150 mg/day	No dosage adjustment		Start low dose – adjust according to response			
Verapamil	**IR:** 40-120 mg q8-12h **SR:** 180-480 q24h	480 mg/day	No dosage adjustment					
Diltiazem	180-480 q24h	480 mg/day	No dosage adjustment		Start low dose			No dosage adjustment
Nifedipine	10-30 mg q8h	90 mg/day	No dosage adjustment	CrCl <20 Start low dose		Start low dose		
Nicardipine	**Initial dose:** 3-5 mg/h x15 min increase by 0.5-1 mg q15min	Max rate: 15 mg/h	No dosage adjustment					
Manidipine	5-20 mg q24h	20 mg/day	No dosage adjustment			No data		
Lercanidipine	5-20 mg q24h	20 mg/day	No dosage adjustment					
Propranolol	80-160 mg bid	320 mg/day	No dosage adjustment					
Atenolol	50-100 mg q24h	100 mg/day	No dosage adjustment	25-50 mg q24h	25 mg q24h	25 mg q24h วันฟอกไต ให้ยาหลังฟอกไต	25 mg q24h	25-50 mg q24h
Metoprolol	50-400 mg q24h	400 mg/day	No dosage adjustment					
Carvedilol	6.25-25 mg q12-24h	25 mg/day	No dosage adjustment					
Bisoprolol	5-20 mg q24h	20 mg/day	No dosage adjustment					
Sildenafil	PAH: 20 mg tid	60 mg/day	No dosage adjustment					
อ้างอิง: Drug Information Handbook, 25th edition; แนวทางการรักษาโรคความดันโลหิตสูงประเทศไทย ปี 2558; The Renal Drug Handbook, 5th edition								

ยาไขมัน

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			60-90	59-15	<15			
Atorvastatin	10-80 mg OD	80 mg/day	No dosage adjustment					
Rosuvastatin	20 mg OD	20 mg/day	CrCl ≥30 No dosage adjustment		CrCl <30 Not exceed 10 mg/day	Not exceed 10 mg/day		
Simvastatin	10-40 mg OD	40 mg/day	CrCl ≥10 No dosage adjustment		CrCl <10 Dose ≤10 mg/day	Dose ≤10 mg/day		No dosage adjustment
Ezetimibe	10 mg OD	10 mg/day	No dosage adjustment					
Gemfibrozil	600 mg bid	1,200 mg/day	eGFR>50: No dosage adjustment	eGFR<50 initially 900 mg daily		Initially 900 mg daily		
Fenofibrate	ขนาดยาที่เหมาะสมแล้วแต่ชนิดยาและบริษัทที่ผลิต					Avoid		Dose as in eGFR 15-20
Cholestyramine	4-16 g OD or bid	24 g/day	No dosage adjustment					

อ้างอิง:

KDOQI clinical practice guidelines and clinical practice recommendations for diabetes and chronic kidney disease2007

Drug Information Handbook, 25th edition; The Renal Drug Handbook, 5th edition

ยาโรคเก๊าท์

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-30	29-10	<10			
Allopurinol	300-400 mg divide bid or tid	900 mg/day	No dosage adjustment	CrCl 20-10: 200 mg/day	100 mg/day	100 mg วันเว้นวัน สามารถเพิ่มขนาดเป็น 300 mg/day วันฟอกไตให้ยาหลังฟอก	No data	200 mg/day
Colchicine	0.6 mg bid, tid	2.4 mg/day	No dosage adjustment	0.6 mg single dose หลีกเลี่ยง chronic use และควรหยุดยาอย่างน้อย 14 วัน ก่อนเริ่มใช้ยาอีกครั้ง		0.6 mg single dose หลีกเลี่ยง chronic use และควรหยุดยาอย่างน้อย 14 วันก่อนเริ่มใช้ยาอีกครั้ง		
Febuxostat	Initial 40 mg OD, maybe increase dose to 80 mg OD	120 mg/day	No dosage adjustment	Start 40 mg and monitor closely		Start 40 mg and monitor closely		

อ้างอิง:

แนวทางเวชปฏิบัติการดูแลรักษาโรคเกาต์ประเทศไทย ปี 2555

Drug Information Handbook, 25th edition; The Renal Drug Handbook, 5th edition

ยารักษาไทรอยด์

ชื่อยา	Normal dose	Max dose	CrCl (ml/min)			Hemodialysis	CAPD	CRRT
			90-60	59-15	<15			
Levothyroxine	100-125 mcg OD	300 mcg/day	No dosage adjustment					
Methimazole	5-15 mg OD	20 mg/day	No dosage adjustment					
Propylthiouracil	50-300 mg divide tid	400 mg/day	No dosage adjustment (renally impaired patient are at greater risk of cardiotoxicity and leucopenia, CrCl <20 ml/min: use with caution)				Use with caution	
อ้างอิง: Drug Information Handbook, 25th edition; The Renal Drug Handbook, 5th edition								

ยาระบบเลือด

ชื่อยา	Normal dose	CrCl (ml/min)				Hemodialysis	CAPD	CRRT
		>50-90	30-50	15-29	<15			
Antiplatelets								
Aspirin	**ACS**; 81-325 mg/day **Analgesic, Antipyretic**; 325-650 mg q4h **1st-2nd prevention**; 75-162 mg/day	No dosage adjustment			CrCl<10 Avoid use	วันฟอกไตให้ยาหลังฟอกไต		No dosage adjustment
Cilostazol	100 mg po q12h ac หรือ pc 2 ชม.	No dosage adjustment						
Clopidogrel	LD 300 หรือ 600 mg then 75 mg/day	No dosage adjustment						
Ticagrelor	LD 180 mg then 90 mg po BID	No dosage adjustment						
Anticoagulants								
Apixaban	**Stroke prophylaxis with AF** 5 mg po BID	No dosage adjustment *แต่ลดขนาดเหลือ 2.5 mg po BID ในผู้ที่มีลักษณะต่อไปนี้ ≥2 ใน 3 ข้อ*: **(1)** age ≥ 80 ปี, **(2)** Wt≤ 60 kg, **(3)** SCr≥ 1.5 mg/dl							
Dabigatran	**Stroke prophylaxis with AF** 110 mg po BID	No dosage adjustment		Avoid				
Rivaroxaban	**Stroke prophylaxis with AF** 20 mg OD with evening meal	No dosage adjustment	15 mg OD		Avoid	Avoid		15 mg OD
Enoxaparin	1 mg/kg SC q12h **Acute DVT;** 1 mg/kg SC q12h หรือ 1.5 mg/kg SC OD	No dosage adjustment		CrCl 15-29 1 mg/kg SC OD	CrCl<15 1 mg/kg SC OD (monitor closely)	Avoid use: if necessary 1 mg/kg SC OD (with monitor closely)		1 mg/kg SC OD
Heparin	1-20,000 unit/mL Adjust dose base on result of aPTT	No dosage adjustment						
Warfarin	Adjust dose base on result of INR	No dosage adjustment						
อ้างอิง: Drug information handbook 25th edition; The Renal Drug Handbook, 5th edition								